# Chapter 12 Review

*Campbell Biology, AP Edition, 12e*

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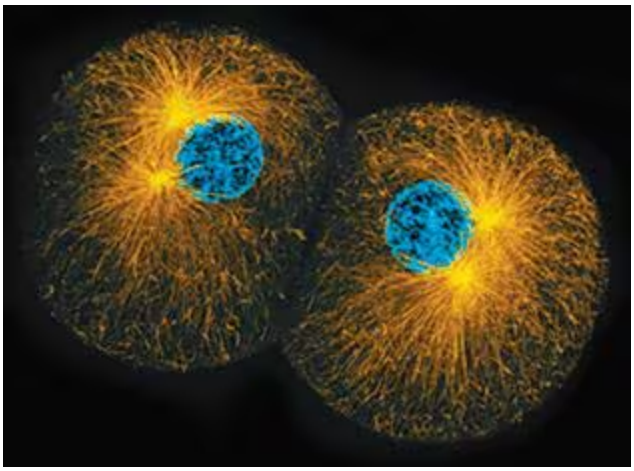


## Summary of Key Concepts

Cells undergoing mitosis in a eukaryote go through a highly conserved and controlled process. There are three major checkpoints in the control system, found in G1, G2, and M phases in a normal cell cycle. However, there are cancer cells that do not follow the same controls. What is out of control in cancer cells? **(BIG IDEA 3)**

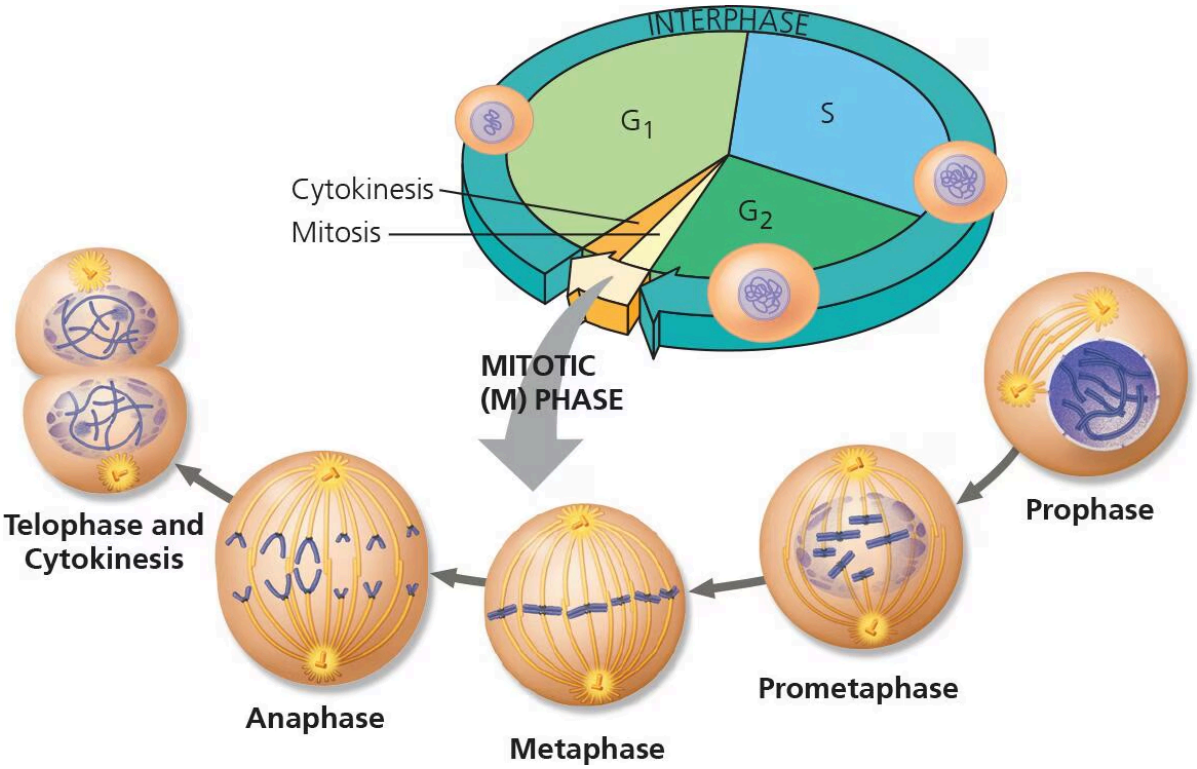
## Concept 12.1: Most cell division results in genetically identical daughter cells

- Unicellular organisms reproduce by **cell division**; multicellular organisms depend on cell division for their development from a fertilized egg and for growth and repair.
- The genetic material (DNA) of a cell—its **genome**—is partitioned among **chromosomes**. Each eukaryotic chromosome consists of one DNA molecule associated with many proteins. Together, the complex of DNA and associated proteins is called **chromatin**. The chromatin of a chromosome exists in different states of condensation at different times. In animals, **gametes** have one set of chromosomes and **somatic cells** have two sets.
- Cells replicate their genetic material before they divide, each daughter cell receiving a copy of the DNA. Prior to cell division, chromosomes are duplicated. Each one then consists of two identical **sister chromatids** joined along their lengths by sister chromatid cohesion and held most tightly together at a constricted region at the **centromeres**. When this cohesion is broken, the chromatids separate during cell division, becoming the chromosomes of the daughter cells. Eukaryotic cell division consists of **mitosis** (division of the nucleus) and ****cytokinesis** (division of the cytoplasm).

*Differentiate between these terms: chromosome, chromatin, and chromatid.*

## Concept 12.2: The mitotic phase alternates with interphase in the cell cycle

- Cell division is part of the **cell cycle**, an ordered sequence of events in the life of a cell.
- Between divisions, a cell is in **interphase**: the G1, **S**, and G2 phases. The cell grows throughout interphase, with DNA being replicated during the synthesis (S) phase. Mitosis and cytokinesis make up the **mitotic (M) phase** of the cell cycle.




Cell cycle diagram consists of interphase, which is divided into G 1, S, and G 2 phases. G 2 is followed by the M phase and cytokinesis. M is divided into Prophase (chromosomes condense, the spindle is forming) which flows to Prometaphase (chromosomes attaching to spindle fibers and moving towards metaphase plate) which flows to Metaphase (chromosomes lined up at metaphase plate with spindle fibers attached) which flows to anaphase (sister chromatids pulled apart by spindle fibers) which flows to telophase and cytokinesis (two distinct daughter nuclei have formed with a cleavage furrow between what will become the new daughter cells).

- The **mitotic spindle**, made up of microtubules, controls chromosome movement during mitosis. In animal cells, it arises from the **centrosomes** and includes spindle microtubules and asters. Some spindle microtubules attach to the **kinetochores** of chromosomes and move the chromosomes to the **metaphase plate**. After sister chromatids separate, motor proteins move them along kinetochore microtubules toward opposite ends of the cell. The cell elongates when motor proteins push nonkinetochore microtubules from opposite poles away from each other.
- Mitosis is usually followed by cytokinesis. Animal cells carry out cytokinesis by **cleavage**, and plant cells form a **cell plate**.
- During **binary fission** in bacteria, the chromosome replicates and the daughter chromosomes actively move apart. Some of the proteins involved in bacterial binary fission are related to eukaryotic actin and tubulin.

- Since prokaryotes preceded eukaryotes by more than a billion years, it is likely that mitosis evolved from prokaryotic cell division. Certain unicellular eukaryotes exhibit mechanisms of cell division that may be similar to those of ancestors of existing eukaryotes. Such mechanisms might represent intermediate steps in the evolution of mitosis.

*In which of the three phases of interphase and the stages of mitosis do chromosomes exist as single DNA molecules?*

## Concept 12.3: The eukaryotic cell cycle is regulated by a molecular control system

- Signaling molecules present in the cytoplasm regulate progress through the cell cycle.
- In the **cell cycle control system**, cyclic changes in regulatory proteins—including **cyclins** and **cyclin-dependent kinases (Cdks)**—work as a cell cycle clock. The cell cycle stops at specific **checkpoints** until a go-ahead signal is received; important checkpoints occur in G1, G2, and M phases. Internal and external signals control the cell cycle checkpoints via signal transduction pathways. Most cells exhibit **density-dependent inhibition** of cell division as well as **anchorage dependence**.
- Cancer cells elude normal cell cycle regulation and divide unchecked, forming tumors. ****Malignant tumors** invade nearby tissues and can undergo **metastasis**, exporting cancer cells to other sites, where they may form secondary tumors.

*Explain the significance of the*G1*,*G2*, and M checkpoints and the go-ahead signals involved in the cell cycle control system.*

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